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Effect of Electro-acupuncture Stimulation of Ximen (PC4) and Neiguan (PC6) on Remifentanil-induced Breakthrough Pain Following Thoracal Esophagectomy

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Summary: The clinical analgesic effect of electro-acupuncture (EA) stimulation (EAS) on breakthrough pain induced by remifentanil in patients undergoing radical thoracic esophagectomy, and the mechanisms were assessed. Sixty patients (ASAⅢ) scheduled for elective radical esophagectomy were randomized into three groups: group A (control) receiving a general anesthesia only; group B (sham) given EA needles at PC4 (Ximen) and PC6 (Neiguan) but no stimulation; and group C (EAS) electrically given EAS of the ipsilateral PC4 and PC6 throughout the surgery. The EAS consisting of a disperse-dense wave with a low frequency of 2 Hz and a high frequency of 20 Hz, was performed 30 min prior to induction of general anesthesia and continued through the surgery. At the emergence, sufentanil infusion was given for postoperative analgesia with loading dose of 7.5 μg, followed by a continuous infusion of 2.25 μg/h. The patient self-administration of sufentanil was 0.75 μg with a lockout of 15 min as needed. Additional breakthrough pain was treated with dezocine (5 mg) intravenously at the patient's request. Blood samples were collected before (T_1), 2 h (T_2), 24 h (T_3), and 48 h (T_4) after operation to measure the plasma β-EP, PGE_2, and 5-HT. The operative time, the total dose of sufentanil and the dose of self-administration, and the rescue doses of dezocine were recorded. Visual Analogue Scale (VAS) scores at 2, 12, 24 and 48 h postoperatively and the incidence of apnea and severe hypotension were recorded. The results showed that the gender, age, weight, operative time and remifentanil consumption were comparable among 3 groups. Patients in EAS group had the lowest VAS scores postoperatively among the three groups (P<0.05). The total dose of sufentanil was 115±6.0 μg in EAS group, significantly lower than that in control (134.3±5.9 μg) and sham (133.5±7.0 μg) groups. Similarly, the rescue dose of dezocine was the least in EAS group (P<0.05) among the three groups. Plasma β-EP levels in EAS group at T_3 (176.90±45.73) and T_4 (162.96±35.00 pg/mL) were significantly higher than those in control (132.33±36.75 and 128.79±41.24 pg/mL) and sham (136.56±45.80 and 129.85±36.14 pg/mL) groups, P<0.05 for all. EAS could decrease the release of PGE_2. Plasma PGE_2 levels in EAS group at T_2 and T_3 (41±5 and 40±5 pg/mL respectively) were significantly lower than those in control (64±5 and 62±7 pg/mL) and sham (66±6 and 62±6 pg/mL) groups. Plasma 5-HT levels in EAS group at T_2 (133.66±40.85) and T_3 (154.66±52.49 ng/mL) were significantly lower than those in control (168.33±56.94 and 225.28±82.03) and sham (164.54±47.53 and 217.74±76.45 ng/mL) groups. For intra-group comparison, plasma 5-HT and PGE_2 levels in control and sham groups at T_2 and T_3, and β-EP in EAS group at T_3 and T_4 were significantly higher than those at T_1 (P<0.05); PGE_2 and 5-HT levels in EAS group showed no significant difference among the different time points (P>0.05). No apnea or severe hypotension was observed in any group. It was concluded that intraoperative ipsilateral EAS at PC4 and PC6 provides effective postoperative analgesia for patients undergoing radical esophagectomy with remifentanil anesthesia and significantly decrease requirement for parental narcotics. The underlying mechanism may be related to stimulation of the release of endogenous β-EP and inhibition of inflammatory mediators (5-HT and PGE_2).
Key words: electro-acupuncture; anesthesia; pain; remifentanil; hyperalgesia

Thoracotomy causes the most severe postoperative pain, and numerous postoperative morbidities, adverse outcomes, and highest incidence of chronic pain[1]. Remifentanil is widely used in general anesthesia because of its rapid onset, strong analgesic effect, short duration, and lack of tendency to accumulate. However, upon discontinuation, breakthrough pain and/or hyperalgesia may occur, necessitating supplemental analgesic administration and decreasing patient's satisfaction with their postoperative analgesic regimen[2]. Although, the parental opioid is the most commonly used practice to treat the postoperative pain, it is limited by its efficacy (e.g. ineffective for cough and ambulation evoked pain and associated side effects for wide application, as well as the risk of respiratory arrest)[3–5].

Thoracic epidural analgesia (TEA) is effective with better analgesia and pulmonary function, and has the highest patient's satisfaction. However, it requires special training, care monitoring, and contraindications in many conditions, such as coagulation disorder, anticoagulation medication, anatomic deformities, infection, and even the refusal by the patient[6–9]. Also, there are

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risks associated with the technique such as dural puncture, neurological injury and paraplegia. Multi-modal approach with local anesthetics infiltration, NSAIDs (such as parecoxib, ketoprofen) have a theoretical advantage[10, 11], but further study is needed to confirm its efficacy and safety in clinical application.

Acupuncture has a long history in treating chronic and acute pain. It has been demonstrated that the EA is also effective adjunct for postoperative pain control[12–15]. Although the exact mechanism is not clear, human and animal studies suggest that the EA produces analgesia through activating the supraspinal descending inhibitory systems, and one of the important components is the endogenous opioids such as enkephalin, endorphin and dynorphin; furthermore using disperse-dense wave of EA can increase the release of 4 types of endogenous opioid peptides[16].

The aim of this study was to prospectively evaluate the analgesic effect of EA acupoints stimulation of Ximen (PC4) and Neiguan (PC6) throughout surgery for postoperative pain. The primary outcome was the effect of EA on postoperative pain score and the difference of opioids doses used among different groups; the second outcome was to explore its effect on plasma β-endorphin (β-EP), prostaglandin E_2 (PGE_2), and serotonin (5-HT).

1 MATERIALS AND METHODS

1.1 General Information

Sixty patients who underwent radical esophagectomy for cancer from June 2012 to June 2013 were included in this study. The study protocol was reviewed and approved by the hospital's ethics committee and all patients were informed and signed the consent form. The patients were randomly divided into 3 groups by computer generated randomization number (n=20 for each group): group A (control) receiving a general anesthesia only; group B (sham) given EA needles at PC4 (Ximen) and PC6 (Neiguan) but no stimulation; group C (EAS) electrically given EAS of the ipsilateral PC4 and PC6 throughout the surgery (detailed description below). The patients' demographic data are shown in table 1. There were 47 men and 13 women, ASAIII, aged 30–70 years old, weighing 50–75 kg. The patients with a history of mental illness, chronic pain, significant chronic cardiac, pulmonary, hepatic, renal or coagulation disorders were excluded from this study. Additionally, those with long-term use of analgesic drugs, any previous surgery or injury, as well as local skin infection occurring in the forearm along the meridian, and a history of previous injuries in the brachial plexus were excluded.

1.2 Anesthesia Methods

All study subjects were premedicated with diazepam (10 mg) and penehyclidine hydrochloride (0.5 mg) intramuscularly 30 min prior to induction of anesthesia. Electrocardiogram was monitored, and an arterial line was placed in the right (left) radial artery for invasive blood pressure monitoring. The general anesthesia was induced by intravenous anesthesia with remifentanil and propofol by a target controlled infusion (TCI) system (CP600TCI, Slgo Technology Co. Ltd, China) to target the plasma concentration at 4 ng/kg, and 3 μg/kg, respectively, and rocuronium (0.8 mg/kg) to facilitate intubation. A left side double lumen endobronchial tube (DLT) was inserted, and confirmed by a fiberscope. All subjects were mechanically ventilated (GE/Datex- Omeda 7900, USA) with volume control node, the tidal volumes were set at 6–8 mL/kg and the adjustments were made to maintain $ETCO_2$ in the range of 35%–45% throughout the surgery. A central vein line was placed in the right jugular vein for fluid administration. One lung ventilation was performed during the surgery to facilitate surgical operation. The anesthetics were maintained by TCI system with the targeted plasma concentrations of 2–4 μg/mL and 2–6 ng/mL for propofol and remifentanil (Batch number: 050501, Renfu Parmaceutical Co. Ltd, China) respectively. Intermittent intravenous boluses of cisatracurium were given for muscle relaxation and monitored by Train of Four (TOF, Watch SX; Organon, Ireland). Bispectral index (BIS) was applied to maintain a value of 45–55 intraoperatively. Sufentanil (Aipeng Epristeride Medical Devices Co., China) infusion was begun 30 min before the conclusion of the surgery and continued up to postoperative day 3 for pain control. A loading dose of sufentanil 7.5 μg was given followed by continuous infusion of 2.5 μg/h. At the end of surgery and the return of spontaneous respiration, the muscle relaxation was reversed, and the patients were extubated and transported to anesthesia recovery room.

1.3 EAS

Based on previous studies[17], the ipsilateral acupoints of Ximen (PC4) and Neiguan (PC6) were chosen for perioperative stimulation. Neiguan (PC6) is located at 6.5 cm proximal to the volar wrist stripes in the midline of forearm and between the tendon of flexor carpi radialis and tendon palmaris longus. Ximen (PC4) is 16.5 cm to the proximal midpoint of the volar wrist stripes (fig. 1). The filiform acupuncture needles (0.38 mm×40 mm) (Huato, Suzhou Medical Supplies Co. Ltd, China) were used. The needle was perpendicularly placed into the skin (1–2 cm) and the patient reported a sensory of soar/expending. EAS was done 30 min before the induction of general anesthesia, and continued throughout the entire surgery. The electric stimulation consisted of a disperse-dense wave with a low frequency of 2 Hz and a high frequency of 20 Hz with a HANS acupuncture instrument (HANS-200, Nanjing Jisheng Medical Science and Technology Ltd, China). The intensity of the stimulation was adjusted to the submaximal level that could be tolerated by the patient.

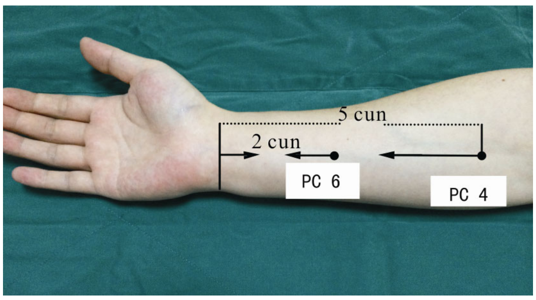

Fig. 1 Anatomical positions for acupoints Ximen (PC4) and Neiguan (PC6)
1 cun=3.33 cm

1.4 Postoperative Pain Assessment and Total Amount of Sufentanil Consumption

Visual analgesia scale (VAS) was used by asking the patient to read a 10-cm scaled roller, 0 for no pain and 10 for the worst pain, which included both the pain in rest and during cough or ambulation postoperatively at 2, 12, 24, and 48 h. The patient controlled intravenous analgesia (PCIA) would allow a bolus of 0.75 μg sufentanil every 15 min for increased pain by the patient. The total amount of sufentanil was recorded, as well as the incidence of apnea, hypotension [mean arterial pressure (MAP) <20%–30% of the baseline blood pressure or systolic blood pressure <90 mmHg]. The breakthrough pain was treated with dezocine (5.0 mg) intravenously by a nurse at the patient's request.

1.5 Determination of Plasma β-EP, 5-HT, and PGE_2

The whole blood samples (2 mL) were collected at four time points from the central venous line: T1, prior to induction; T2, T3, and T4, 2, 24, and 48 h postoperation, respectively. The blood (2.0 mL) was added into a vial containing 10% EDTA and aprotinin, and placed at 4°C for 15 min, then centrifuged at 3500 r/min. The plasma was transferred to a separate vial and stored at –70°C until analysis. Radioimmunoassay was used to determine plasma concentrations of β-EP and PGE_2. ELISA was used to determine plasma concentration of 5-HT. Assay kits were obtained from Beijing Huaying Biotechnology Research Institute (China). The persons who collected the data and performed the data analysis were blinded from the group and patients assignment.

1.6 Statistical Analysis

Data were expressed as $\bar{x}$±*s*. SPSS13.0 (SPSS for Windows, version 13.0; SPSS Inc., USA) was used for data analysis. Comparison of the patient's demographics and the amount of anesthetic in three groups were performed by one-way analysis of variance for continuous data and chi-square for categorical data. Post-hoc analyses were performed with the Fisher's LSD test for pairwise comparison between groups. Rank sum testing was performed to compare two sets of count data. The level of significance was set at 0.05 (two-tailed).

2 RESULTS

2.1 General Information

The gender, age, weight, operative time and remifentanil consumption were comparable among 3 groups. There were no significant differences in operating time, the total amount of propofol and remifentanil among 3 groups (P>0.05, table 1).

Table 1 Patient's demographics and the amount of anesthetic in three groups ($\bar{x}$±*s*, *n*=20)

Items	A	B	C
Sex (male/female)	14/6	15/5	14/6
Age (years)	53±8	55±10	54±7
Weight (kg)	59±9	60±8	59±7
Operating time (min)	218±27	221±30	217±33
Propofol (mg)	1048±167	1077±180	1064±174
Remifentanil (mg)	1.90±0.65	1.90±0.59	1.89±0.63

A: control group; B: sham EA group: C: EAS group.

2.2 Total Doses of Sufentanil and Dezocine Used for Treating Postoperative Pain

The total doses of sulfentanil were 115±6.0 μg in EAS group, significantly less than that in control and sham groups (P<0.05, table 2). Patients in control and sham groups had three times more self administration doses (table 2). Similarly, the patients of control and sham groups requested dezocine to treat the breakthrough pain (P>0.05, table 2).

Table 2 The total dose of sufentanil, patient's demand and rate of pain rescue with dezocine ($\bar{x}$±*s*, *n*=20)

Groups	Total dose of sufentanil (μg)	Patient demand	Rate of rescue with dezocine (%)
A	134±5.9	42±5	35
B	133±7.0	43±3	30
C	115±6.0*	16±3*	5*

The total dose of sufentanil, patient's demand and rate of pain rescue with dezocine in group C are lowest among the three groups, $^*P$<0.05.

2.3 Postoperative VAS Scores

The patients in EAS group had the lowest VAS score among the three groups during the entire study period (P<0.05, fig. 2), and they were rest more comfortable and tolerated coughing and ambulation as well as going to toilet. There was no significant difference between control and sham groups in VAS scores (P>0.05, fig. 2). These results pointed out that EAS can reduce the postoperative narcotics and enhance the analgesic effect.

2.4 Plasma β-EP, PGE_2 and 5-HT Levels

Plasma β-EP levels were significantly increased immediately after the surgery in all groups, but elevated only in EAS group at 24 and 48 h postoperatively (176.90±45.7 and 162.96±35.0 pg/mL, respectively compared to the baseline, P<0.05, fig. 3).

Plasma PGE_2 levels were significantly elevated in control and sham groups at 2 and 24 h after the surgery, in contrast, those were significantly reduced in EAS group at the same time points (P<0.05, fig. 4). There was no significant difference at 72 h postoperatively.

There was a sustained increase in plasma 5-HT levels postoperatively up to 72 h in all groups (fig. 5), but 5-HT levels were significantly reduced in EAS group as compared to those in control and sham groups (P<0.05, fig. 5).

3 DISCUSSION

Acupuncture is a well-known complementary treatment for pain with few or no side effects[18]. However its utility in acute pain management began recently and a few studies explored its application for postoperative pain control[19, 20]. This study demonstrated that EAS could be an adjunct effectively managing the post-thoracotomy pain, by which the patients reported less pain and consumed less nacortics postoperatively.

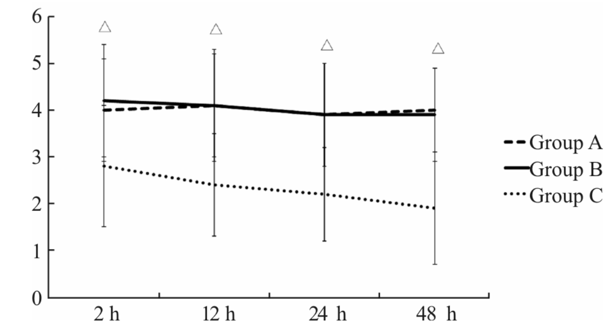

Fig. 2 Postoperative changes of VAS scores in the three groups VAS scores at 2, 12, 24, and 48 h after the operation were lower in group C than in groups A and B ($^{\triangle}P$ <0.05).
Group A: control group; group B: sham EA group; group C: EAS group

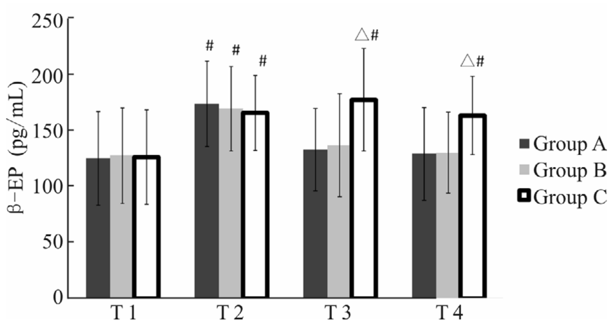

Fig. 3 Plasma β-EP levels were significantly increased immediately after the surgery in all groups, but elevated only in group C at 24 and 48 h postoperatively ($^{\triangle}P$<0.05 *vs*. group A/B; $^{\#}P$<0.05 *vs*. T1).
Group A: control group; group B: sham EA group; group C: EAS group

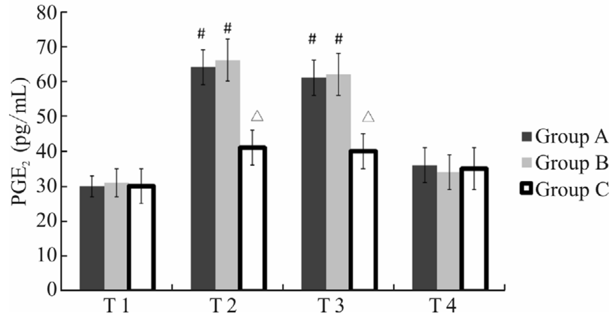

Fig. 4 Plasma PGE_2 levels were significantly elevated in group A and B at 2 and 24 h postoperatively; in contrast, those were significantly lower in group C than in groups A and B at the same time points ($^{\triangle}P$<0.05 *vs*. group A/B; $^{\#}P$<0.05 *vs*. T1).
Group A: control group; group B: sham EA group; group C: EAS group

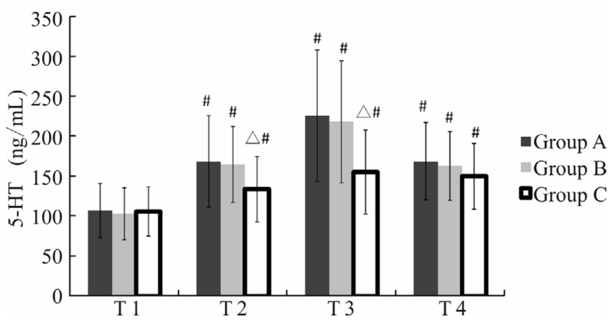

Fig. 5 Plasma 5-HT levels were significantly lower in group than in groups A and B ($^{\triangle}P$<0.05 *vs*. group A/B; $^{\#}P$<0.05 *vs*. T1).
Group A: control group; group B: sham EA group; group C: EAS group

Thoracotomy for lung or esophagus cancer produces the most severe pain postoperatively which also is associated with significant perioperative complications if not treated effectively, and a high incidence to develop into chronic pain[21]. Patients who receive remifentanil for anesthesia after radical esophagectomy will experience breakthrough pain or significant hyperalgesia necessitating the administration of large doses of analgesic agents to produce satisfactory results[22]. The mechanism of hyperalgesia is unclear. It is still a challenge to effectively control such a pain clinically[23]. Although numerous studies suggested the efficacy of thoracic epidural analgesia, its wide application is limited by special training and intensive monitoring and troubleshooting, it would be contraindicated if the patient is on anticoagulation agents, others for local infection, even the patient's refusal; on the other hand, the nacortics either by PCA or nurse administration on demands, provides inferior analgesia compared to thoracic epidural analgesia, always there is a risk of tolerance, overdose, and the risk of respiration depression. Others such as the multimodal approach are currently under investigation. Therefore, the approach with EAS has its unique property in the postoperative pain management. The technique is simple and straightforward, which requires minimal training. As compared to control and sham groups, the patients in EAS group required the less amount of sufentanil up to postoperative 48 h. Being safer, the application of EAS could require less nursing staff's work intensity and improve the patient's satisfactions.

EA has a long history being used in the treatment for various pain conditions, and its efficacy was evaluated extensively both in animal and human experiments[24–26]. In rats, EA can effectively inhibit spontaneous behaviors such as the retraction of the legs and licking of the paws[27, 28]. In humans, EA stimulation can reduce the transformation of noxious stimuli to the central nervous system and alleviate or prevent central sensitization[29, 30]. In addition, the analgesic effect has a frequencies dependent relationship.

Studies have demonstrated that stimulation at PC4 and PC6 could control thoracalgia, angina, and diaphragm spasm[17, 31], which was the reason that PC4 and PC6 were chosen to evaluate their analgesic effect for post-thoracotomy pain. Although the exact mechanisms of acupuncture inducing analgesia are still not clear, several lines of evidence suggest it works by a complex activation and inhibition interaction[32].

β-EPs are neuropeptides involved in pain management, possessing morphine like effects[33]. β-EP is synthesized and released by the hypothalamus which is one of major endogenous pain control systems[34]. In rats, the stress-induced release of endogenous β-EP produces antinociceptive and antihyperalgesic effects[35]. The analgesia effect of acupuncture is mainly mediated through activation of descending inhibitory system including opioidergic, adrenergic and serotonergic pathways at both central and peripheral nervous systems[36, 37]. In current study, the additional β-EP could be released from macrophage by EAS and surgery, and enhance the EAS analgesia postoperatively.

The release of proinflammatory mediators and cytokines plays an important role in sustained hyperalgesia[38, 39]. Postoperative pain/hyperalgesia is directly related to the extensiveness of surgical tissue injury. Our study clearly showed there was a significant increase in plasma PGE_2 and 5-HT levels. PGE_2, a potent inflammatory mediator, produces pain directly and increases the nociceptor's sensitivity to algogenic substances. EA has been shown to inhibit COX expression in the dorsal horn of the spinal cord, reduce the release of inflammatory factors and significantly inhibit hyperalgesia. 5-HT is a classic pain-related substance. Studies have shown that the 5-HT1A receptor subtype is involved in 5-HT-mediated mechanical hyperalgesia, and 5-HT2A participates in hyperalgesia after inflammation caused by subcutaneous injection of Freund's adjuvant[40].

In summary, the present study suggests intraoperative ipsilateral EAS at PC4 and PC6 as an adjunct could provide effective postoperative analgesia for patients undergoing radical esophagectomy with remifentanil anesthesia and significantly decrease requirement for parental narcotics. The underlying mechanism may be related to stimulation of the release of endogenous β-EP and inhibition of the inflammatory mediators (5-HT and PGE_2).

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Conflict of Interest Statement

The authors declare that there is no conflict of interest with any financial organizations or corporations or individuals that can inappropriately influence this work.

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